

Wuao Liu

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EDUCATION	University of Massachusetts Amherst <i>Ph.D. in Computer Science</i> University of Michigan <i>M.S. in Robotics</i> Zhejiang University <i>B.Eng. in Robotics</i>	Aug. 2024 – Apr. 2028 (expected) Amherst, MA Aug. 2021 – Apr. 2023 Ann Arbor, MI Sep. 2017 – Jun. 2021 Hangzhou, China
PUBLICATIONS	- Mustafa Chasmai, Wuao Liu, Subhransu Maji, and Grant Van Horn. Audio Geolocation: An Investigation with Natural Sounds. <i>Under Review</i>. [paper] - Filippos Bellos, Yayuan Li, Wuao Liu, and Jason Corso. Can Large Language Models Reason About Goal-Oriented Tasks? In <i>Proceedings of ACL Workshop on the Scaling Behavior of Large Language Models</i>, pages 24-34, 2024. [paper]	
WORK EXPERIENCE	Honda Research Institute USA, Inc. <i>Research Scientist Intern</i> - Presented a novel framework that integrates first-order ambisonics spatial audio cues with LLMs, enabling advanced spatial audio understanding for wearable applications. - Designed a spatial audio encoder optimized for sound recognition, direction-of-arrival (DoA) estimation, and spatial reasoning. - Introduced the FoASoundQA dataset, comprising 872k multimodal questions covering both perceptual and reasoning tasks, and benchmarked performance against AudioLLMs, including OpenAI Whisper, SALMONN, and Qwen2-Audio-7B. University of Michigan <i>Research Engineer</i> - Developed a perception system for AR glasses to assist users with cooking tasks. [project] - Designed and optimized a data collection protocol, resulting in a curated dataset of 224 instructional videos. - Implemented a Faster R-CNN object detector to identify 14 kitchen items, achieving a mean Average Precision (mAP) of 85.99 on the curated dataset. - Implemented a novel transformer-based procedural step recognition model, achieving 92% accuracy and enabling automated analysis of step transitions errors.	May. 2025 – Aug. 2025 San Jose, CA Jun. 2023 – Apr. 2024 Ann Arbor, MI
RESEARCH EXPERIENCE	University of Massachusetts Amherst Advisor: Dr. Grant Van Horn Project: Weakly-supervised Learning for Natural Audio - Extended Audio Masked Autoencoder (AudioMAE) pretraining framework to wildlife audio, enabling self-supervised representation learning via spectrogram reconstruction. - Achieved a 1.6% improvement in sound classification accuracy on iNatSounds dataset. University of Massachusetts Amherst Advisor: Dr. Grant Van Horn Project: Contrastive Learning of Audio and Geographical Data - Proposed the first framework for geolocating natural audio at a global scale. - Developed a retrieval-based approach that integrates audio geolocation and species range prediction, achieving a median localization error of 1127km and a regional-level localization accuracy of 17.2%.	Jan. 2025 – Apr. 2025 Amherst, MA Aug. 2024 – Dec. 2024 Amherst, MA
TECHNICAL SKILLS	- Programming Languages: Python, C / C++, MATLAB, L^AT_EX. - Deep Learning Frameworks: PyTorch, TensorFlow, Hugging Face. - Machine Learning Packages: Scikit-learn, JSON, Pandas	